

Mealyzer Protocol for Dietitians

I. Introduction

Welcome to Mealyzer! The goal of this application is to help participants with DM2 learn to better assess the nutritional content of their meals.

Registered dietitian/nutritionists (RDNs) will provide macronutrient, fiber, and energy estimates of meals based on images and brief descriptions the users provide.

This protocol is designed to enhance inter-rater reliability among study dietitians by setting specific guidelines to be followed when completing each meal analysis.

II. Estimating Portion Sizes

- When you open a nutritional evaluation request you will see a photo of the meal and a brief description of its contents. The user may also provide you with estimated amounts of each food or ingredient. If these amounts seem reasonably accurate, then use them as the basis of your assessment. If, however, the user's estimates do not appear accurate, you may use your expertise to adjust the amounts accordingly.
- When estimating portion sizes, you may wish to consult the examples illustrated in the Serving Size Booklet:
<https://www.fredhutch.org/content/dam/www/research/divisions/public-health-sciences/nutrition-assessment/serving-size-booklet-sample.pdf>

III. Estimating Foods that Cannot be Visualized and/or Have Incomplete Descriptions

- **Hidden foods** - Some images may contain foods or ingredients that cannot be visualized. Examples include milk or sugar added to coffee, fat used for cooking, or dressing tossed in a salad. In these instances, you should rely first on the amounts listed in the user's description.
- **Hidden foods with no description** - If the user's description does not contain amounts for unseen ingredients, please use your expertise to estimate how much of the ingredients would likely be used. For example, if the user states that a sandwich contains mayonnaise, but they do not list an amount and you cannot see it, then 1 TBSP of mayonnaise would be a reasonable guess.
- **Note on alcoholic beverages:** if the user states that they consumed an alcoholic beverage but did not include it in the photo and did not describe the portion size, base the assessment on a standard serving of that beverage.
Standard serving sizes for alcoholic beverages can be found here: <http://www.niaaa.nih.gov/alcohol-health/overview-alcohol-consumption/what-standard-drink>
- **Note on milk products:** in the case that a user submits an image containing a milk product and does not state whether the product is full-, low-, or non-fat, base the

assessment on the **low-fat or 2%** option for that product.

IV. Retrieving Nutrition Information

- **USDA FoodData Central** is the preferred database for retrieving nutrition information for the purposes of meal analysis in Mealyzer. Dietitians should make every effort to obtain nutrition information here before searching alternate resources. The database can be accessed here: <https://fdc.nal.usda.gov/index.html>
- **The ADA/AND Diabetic Exchange List** – for some mixed dishes, it may be necessary and/or significantly faster to use the exchange list as the bases for your assessment. Examples include casseroles, soups/stews, or mixed vegetable dishes where it is difficult to break down amounts of individual ingredients. In these cases, you may refer to the Choose Your Foods: Food lists for Diabetes or other exchange list booklet published by the Academy of Nutrition and Dietetics.

The downside to using the exchange system is that you will not be able to give an estimate for fiber. In these cases, RDNs must use their skills to make an educated guess based on the foods present in the dish (for example, beans in chili will increase fiber content). Remember to check the USDA Nutrient Database first before using the exchange list. You may be able to find similar items there.

- **Free Foods** - For the purposes of making assessments more time efficient, RDNs have the discretion to exclude foods that do not contribute significant nutrients to the meal. Examples include lettuce and

tomato on a hamburger or other garnishes added to a plate. As a rule of thumb, free foods are foods that contribute less than 20 Calories and 5 grams of carbohydrate to a meal. If multiple free foods are included in a single meal assessment, the total nutrient contribution that would be excluded through free foods should not exceed 20 Calories or 5 grams of carbohydrate.

V. **Sample Evaluation**

Description: Whole grain toast, 2 TBSP almond butter, $\frac{1}{2}$ apple, tea with 2 sugars

Ingredients: Whole wheat bread, almond butter, apple, tea, sugar



	CHO (g)	Protein (g)	Fat (g)	Fiber (g)	Kcal (g)
1 slice whole grain bread (32 g)	13.7	3.97	1.12	1.92	80.6
2 TBSP Almond Butter	6.02	6.72	17.76	3.3	196.4
½ a large apple, raw with skin	15.4	0.29	0.19	2.68	58
2 teaspoons sugar	7.96	0	0	0	32
Tea	free food	free food	free food	free food	free food

Totals: Carbs = 43 grams, Protein = 11 grams, Fat = 19 grams, Fiber = 8 grams, Calories = 367 kcal

References:

1. Sample Serving Size Booklet. The Fred Hutchinson Cancer Research Center Website:
<https://www.fredhutch.org/content/dam/www/research/divisions/public-health-sciences/nutrition-assessment/serving-size-booklet-sample.pdf>
Accessed January 13, 2022
2. What is a Standard Drink? The National Institute on Alcohol Abuse and Alcoholism Website:
<http://www.niaaa.nih.gov/alcohol-health/overview-alcohol-consumption/what-standard-drink>
Accessed January 13, 2022
3. The USDA Nutrient Database. The USDA Website:
<https://fdc.nal.usda.gov/index.html>
Accessed January 17, 2022.